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Ex.No 10: 5G Network Slicing Environment Using MATLAB

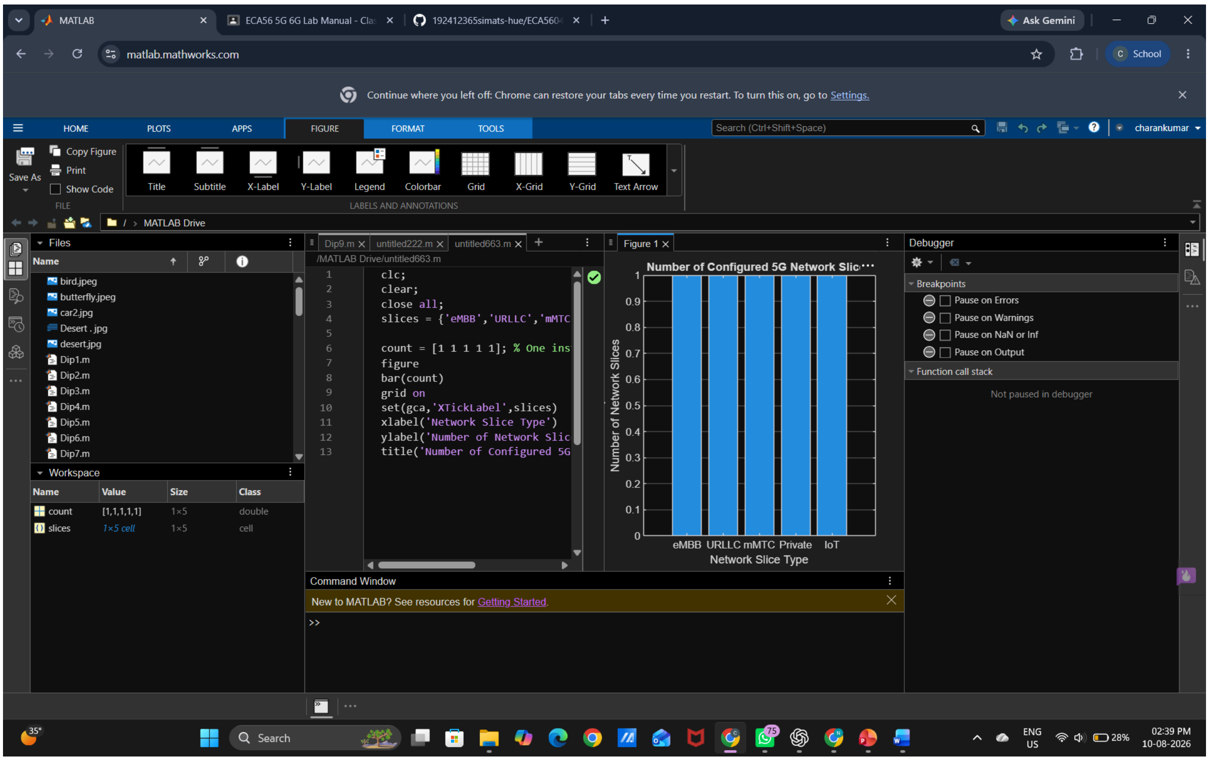
Objective 1:

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Program (Matlab Code)

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Output:



Objective 2:

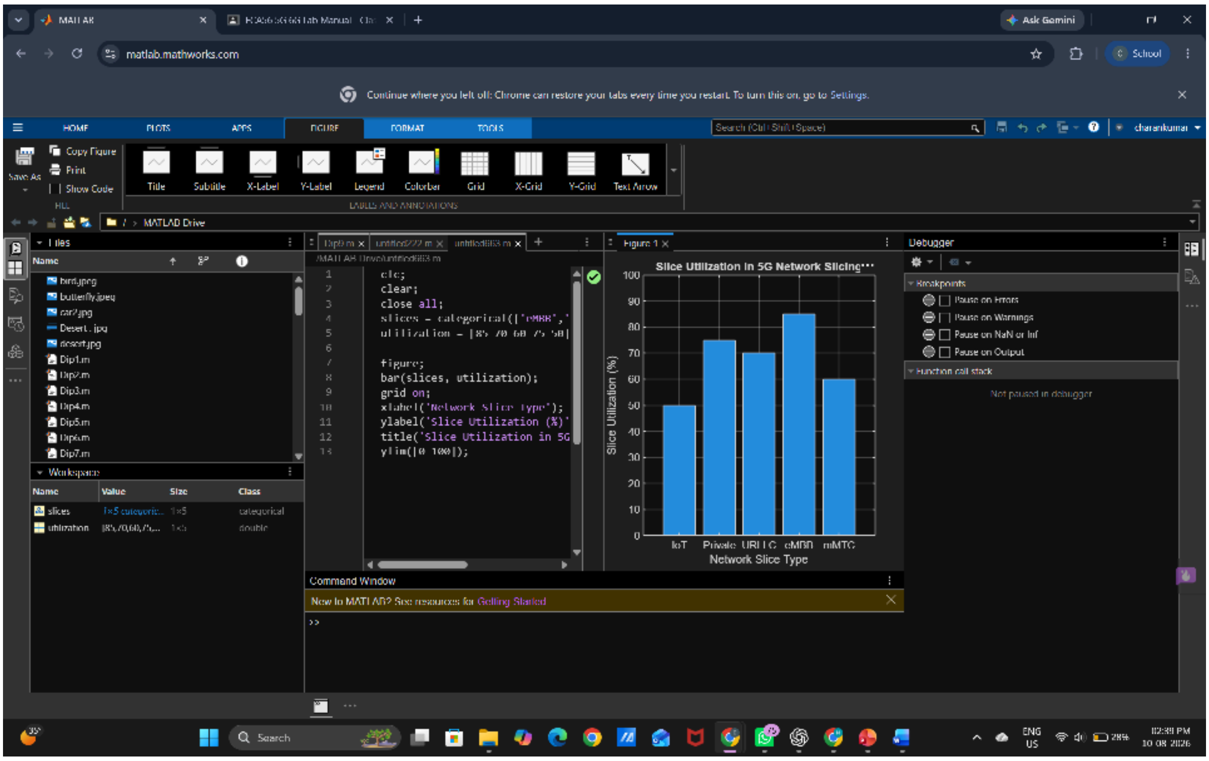
Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Matlab Code

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Output



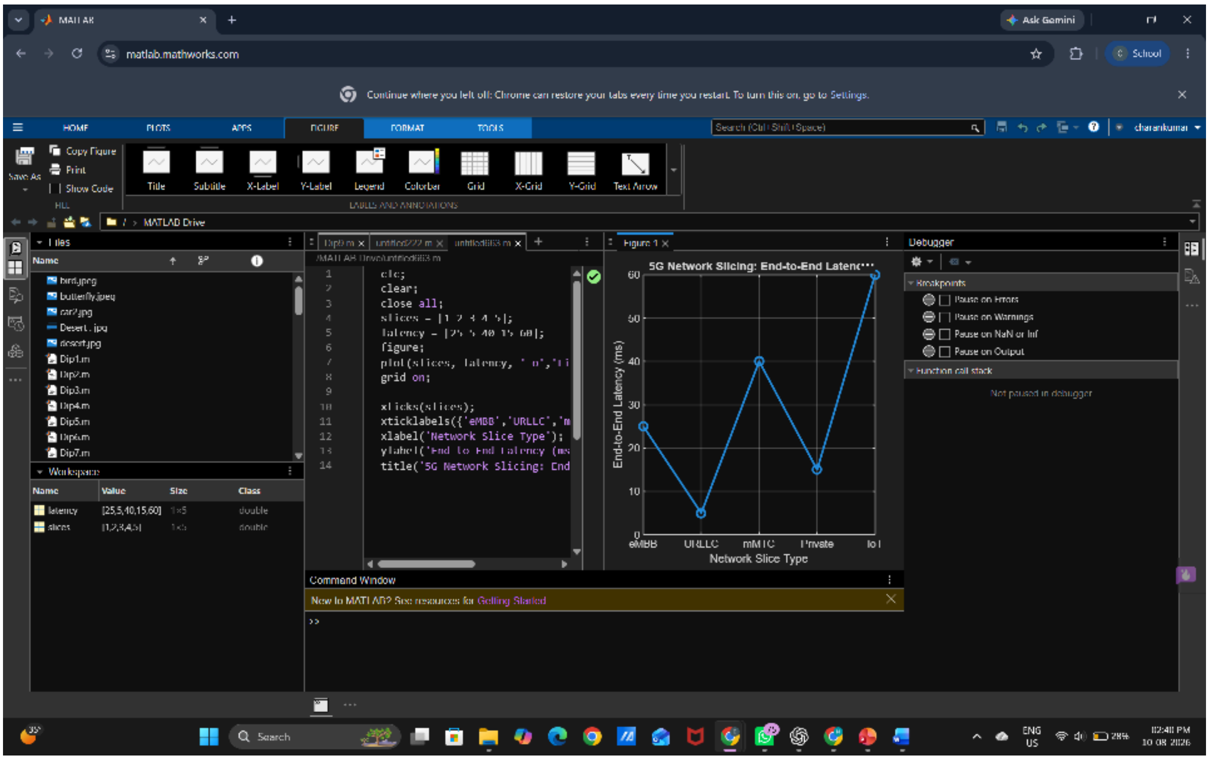
Objective 3:

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

Matlab Code

```
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```

Output



matlab.mathworks.com
clc;
clear;
close all;
figure;
grid on;
xticklabels({'eMBB','URLLC','m
xlabel('Network Slice Type');
title('5G Network Slicing: End
5G Network Slicing: End-to-End Latenc
End-to-End Latency (ms)
eMBB
URLLC
Network Slice Type
Command Window